

Challenge (Habanero)

- Q1.** A recipe for making cookies requires $2\frac{1}{3}$ cups of flour. If the bakery has 15 cups of flour, how many batches can they make?
(A) 3
(B) 5
(C) 2
(D) 1
(E) 4
- Q2.** A construction company has 5 workers, and their daily working hours are 8, 10, 7, 9, 8. What is the mode of their working hours?
(A) 7
(B) 8
(C) 9
(D) 10
(E) 5
- Q3.** A set of architectural plans has the following dimensions in inches: 10, 12, 15, 11, 13. What is the median of these dimensions?
(A) 11
(B) 12
(C) 13
(D) 10
(E) 15
- Q4.** The mean of 5 test scores is 80. If one score is 70, what is the sum of the other 4 scores?
(A) 300
(B) 280
(C) 320
(D) 310
- (E) 350
- Q5.** The median of 3 numbers is 7. If the numbers are 5, 7, and x , what is the value of x ?
(A) 5
(B) 7
(C) 9
(D) 3
(E) 10
- Q6.** The mode of the following set of numbers is 4: 2, 4, 6, 4, 8. What percentage of the numbers is 4?
(A) 20
(B) 40
(C) 60
(D) 50
(E) 30
- Q7.** The mean of two numbers is 9, and their difference is 3. What are the numbers?
(A) 6, 12
(B) 3, 6
(C) 6, 3
(D) 12, 6
(E) 9, 3
- Q8.** A set of measurements has the following values: 10, 12, 14, 16, 18. What is the range of these measurements?
(A) 4
(B) 8
(C) 16
(D) 10
(E) 6

Open Ended Questions (FRQ)

- Q9.** A bakery is making a special batch of bread. The recipe requires 2 cups of flour per loaf. The bakery has 25 cups of flour and wants to make as many loaves as possible. Calculate the maximum number of loaves they can make and explain your reasoning.
- Q10.** A construction company is building a new high-rise. The blueprint has the following dimensions: length = 50 meters, width = 20 meters. Calculate the area of the building and explain how you would use the mean, median, and mode to analyze the dimensions of the building.

Chef's Tips

- Be sure to check for unit consistency in calculations.
- Encourage students to use real-world applications to understand concepts.
- Ensure students show all work and explain reasoning for free response questions.